AERIS Expert Review Packet

DRAFT

Anomaly

Source / metric	openaq / ozone
Observed / expected	0.04 / 0.0147256
Time	2026-07-08 03:00 UTC
Location	29.5203, -95.3925
Severity	moderate
Detectors	stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	7 / 704	39.12 to 100; mean 73.2391	64.96 at 2026-07-08 02:55Z; dt 5 min
asos	temperature	degC	7 / 704	23 to 36.1111; mean 28.9822	26.1111 at 2026-07-08 02:55Z; dt 5 min
asos	wind_direction	degree	7 / 854	0 to 340; mean 136.897	60 at 2026-07-08 02:55Z; dt 5 min
asos	wind_speed	m/s	7 / 889	0 to 10.8033; mean 2.3529	3.08666 at 2026-07-08 02:55Z; dt 5 min
noaa_gfs	gh_500	m	8 / 96	5896.24 to 5924.75; mean 5910.78	5911.85 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	pbl_height	m	8 / 96	19.6394 to 2153.73; mean 722.402	1003.53 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	precipitable_water	mm	8 / 96	39.1305 to 51.5499; mean 46.114	49.6923 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	surface_pressure	hPa	8 / 96	1008.95 to 1017.68; mean 1014.36	1012.9 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	t_850	degC	8 / 96	17.4065 to 20.4926; mean 19.0548	18.8821 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	u_10m	m/s	8 / 96	-2.34889 to 1.76546; mean 0.2332	-1.33889 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	v_10m	m/s	8 / 96	-0.854414 to 4.60966; mean 2.2122	4.3194 at 2026-07-08 00:00Z; dt 180 min
openaq	ozone	ppm	16 / 726	-0.004 to 0.058; mean 0.017	0.04 at 2026-07-08 03:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 145	8 to 75; mean 25.5793	8 at 2026-07-08 03:00Z; dt 0 min
openaq	pm25	ug/m3	77 / 2668	0 to 62.1; mean 5.3824	4.8 at 2026-07-08 03:00Z; dt 0 min
openweather	cloud_cover	percent	4 / 287	0 to 100; mean 23.0209	24 at 2026-07-08 03:02Z; dt 2 min
openweather	humidity	percent	4 / 287	38 to 95; mean 76.2056	84 at 2026-07-08 03:02Z; dt 2 min
openweather	precipitation	mm/h	4 / 287	0 to 5.62; mean 0.0572	0 at 2026-07-08 03:02Z; dt 2 min
openweather	pressure	hPa	4 / 287	1012 to 1019; mean 1016.36	1017 at 2026-07-08 03:02Z; dt 2 min
openweather	temperature	degC	4 / 287	23.78 to 35.9; mean 29.5798	26 at 2026-07-08 03:02Z; dt 2 min
openweather	wind_direction	degree	4 / 287	0 to 359; mean 193.801	173 at 2026-07-08 03:02Z; dt 2 min
openweather	wind_speed	m/s	4 / 287	0.36 to 8.39; mean 2.2718	3.23 at 2026-07-08 03:02Z; dt 2 min
purpleair	pm25	ug/m3	69 / 4787	0 to 146.4; mean 7.2753	5.9 at 2026-07-08 03:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0243723 to 0.0253077; mean 0.0249	0.0249873 at 2026-07-07 18:57Z; dt 482.4 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_hcho_column	mol/m^2	8 / 8	0.000174097 to 0.000235341; mean 0.0002	0.000202795 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_hcho_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	3.63106e-05 to 4.30727e-05; mean 0	4.10314e-05 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_so2_column	mol/m^2	8 / 8	-3.49304e-05 to 4.72775e-05; mean 0	2.51087e-05 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_so2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
tceq	co	ppm	3 / 162	0 to 1; mean 0.2383	0.3 at 2026-07-08 03:00Z; dt 0 min
tceq	no2	ppb	12 / 633	-0.1 to 35.8; mean 6.4193	2.9 at 2026-07-08 03:00Z; dt 0 min
tceq	so2	ppb	3 / 154	-0.5 to 6.8; mean -0.0513	-0.4 at 2026-07-08 02:00Z; dt 60 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-06 12Z 60.58, 18Z 55.4, 07-07 00Z 76.95, 06Z 88.92, 12Z 70.38, 18Z 54.05, 07-08 00Z 79.03, 06Z 91.72, 12Z 70.29, 18Z 59.21, 07-09 00Z 77.26, 06Z 89.03, 12Z 78.52
asos	temperature	07-06 12Z 32.2, 18Z 32.8, 07-07 00Z 28.42, 06Z 25.86, 12Z 29.92, 18Z 32.92, 07-08 00Z 26.11, 06Z 24.66, 12Z 30.43, 18Z 31.77, 07-09 00Z 28.21, 06Z 25.65, 12Z 28.62
asos	wind_direction	07-06 12Z 213.9, 18Z 170.4, 07-07 00Z 138.6, 06Z 68.49, 12Z 126.7, 18Z 168.2, 07-08 00Z 116.8, 06Z 29.57, 12Z 186.9, 18Z 169.7, 07-09 00Z 153.7, 06Z 124.1, 12Z 157.6
asos	wind_speed	07-06 12Z 2.322, 18Z 4.208, 07-07 00Z 2.302, 06Z 0.792, 12Z 1.419, 18Z 3.732, 07-08 00Z 2.822, 06Z 0.3963, 12Z 2.178, 18Z 3.995, 07-09 00Z 2.746, 06Z 1.467, 12Z 2.017
noaa_gfs	gh_500	07-06 18Z 5903, 07-07 00Z 5898, 06Z 5905, 12Z 5899, 18Z 5915, 07-08 00Z 5912, 06Z 5918, 12Z 5912, 18Z 5924, 07-09 00Z 5914, 06Z 5919, 12Z 5910
noaa_gfs	pbl_height	07-06 18Z 1786, 07-07 00Z 881.5, 06Z 190.8, 12Z 93.56, 18Z 1918, 07-08 00Z 932.5, 06Z 222.7, 12Z 127, 18Z 1912, 07-09 00Z 211.4, 06Z 264.7, 12Z 128
noaa_gfs	precipitable_water	07-06 18Z 43.14, 07-07 00Z 46.35, 06Z 45.04, 12Z 46.2, 18Z 48.95, 07-08 00Z 49.03, 06Z 46.85, 12Z 46.51, 18Z 48.52, 07-09 00Z 44.83, 06Z 43.93, 12Z 44.03
noaa_gfs	surface_pressure	07-06 18Z 1013, 07-07 00Z 1011, 06Z 1014, 12Z 1015, 18Z 1015, 07-08 00Z 1013, 06Z 1016, 12Z 1016, 18Z 1016, 07-09 00Z 1014, 06Z 1015, 12Z 1015
noaa_gfs	t_850	07-06 18Z 17.9, 07-07 00Z 19.9, 06Z 19.87, 12Z 18.78, 18Z 17.79, 07-08 00Z 19.15, 06Z 19.75, 12Z 19.15, 18Z 17.93, 07-09 00Z 19.91, 06Z 19.47, 12Z 19.05
noaa_gfs	u_10m	07-06 18Z 0.4423, 07-07 00Z -0.9439, 06Z 0.348, 12Z 0.9233, 18Z -0.3567, 07-08 00Z -1.305, 06Z 0.2531, 12Z 1.015, 18Z 0.6143, 07-09 00Z -0.1454, 06Z 0.8586, 12Z 1.094

Source	Metric	Values
noaa_gfs	v_10m	07-06 18Z 1.698, 07-07 00Z 4.085, 06Z 2.138, 12Z 1.049, 18Z 1.598, 07-08 00Z 4.091, 06Z 2.428, 12Z 1.182, 18Z 0.9193, 07-09 00Z 3.199, 06Z 2.689, 12Z 1.469
openaq	ozone	07-06 12Z 0.01926, 18Z 0.03261, 07-07 00Z 0.01312, 06Z 0.005071, 12Z 0.01228, 18Z 0.02885, 07-08 00Z 0.0252, 06Z 0.004012, 12Z 0.004032, 18Z 0.035, 07-09 12Z 0.01162
openaq	pm10	07-06 12Z 33.67, 18Z 27.5, 07-07 00Z 15.89, 06Z 18.28, 12Z 33.17, 18Z 29.88, 07-08 00Z 21.94, 06Z 24.39, 12Z 36, 18Z 23, 07-09 12Z 29.5
openaq	pm25	07-06 12Z 5.905, 18Z 5.42, 07-07 00Z 4.653, 06Z 6.059, 12Z 6.511, 18Z 6.962, 07-08 00Z 4.573, 06Z 6.678, 12Z 5.807, 18Z 3.143, 07-09 00Z 3.655, 06Z 4.584, 12Z 5.286
openweather	cloud_cover	07-06 12Z 13.73, 18Z 25.67, 07-07 00Z 5.44, 06Z 2.167, 12Z 25.67, 18Z 47.04, 07-08 00Z 29.54, 06Z 4.208, 12Z 3.913, 18Z 31.57, 07-09 00Z 68.88, 06Z 21.17, 12Z 4.25
openweather	humidity	07-06 12Z 69.55, 18Z 60.88, 07-07 00Z 75.8, 06Z 87.42, 12Z 76.67, 18Z 60.25, 07-08 00Z 78.04, 06Z 91.54, 12Z 79.39, 18Z 61.87, 07-09 00Z 75.56, 06Z 88.38, 12Z 86.42
openweather	precipitation	07-06 12Z 0, 18Z 0.006667, 07-07 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-08 00Z 0.6458, 06Z 0, 12Z 0, 18Z 0.03261, 07-09 00Z 0, 06Z 0, 12Z 0
openweather	pressure	07-06 12Z 1016, 18Z 1014, 07-07 00Z 1014, 06Z 1016, 12Z 1018, 18Z 1016, 07-08 00Z 1016, 06Z 1017, 12Z 1019, 18Z 1017, 07-09 00Z 1016, 06Z 1016, 12Z 1018
openweather	temperature	07-06 12Z 32.17, 18Z 34.28, 07-07 00Z 29.56, 06Z 26.44, 12Z 30.09, 18Z 34.14, 07-08 00Z 27.75, 06Z 24.6, 12Z 29.73, 18Z 33.33, 07-09 00Z 29.16, 06Z 26.26, 12Z 27.63
openweather	wind_direction	07-06 12Z 216.9, 18Z 182.6, 07-07 00Z 190.7, 06Z 210.5, 12Z 196.1, 18Z 167.8, 07-08 00Z 169.3, 06Z 193, 12Z 231.1, 18Z 185, 07-09 00Z 176.8, 06Z 206.2, 12Z 222
openweather	wind_speed	07-06 12Z 1.665, 18Z 2.22, 07-07 00Z 2.398, 06Z 1.865, 12Z 1.447, 18Z 2.462, 07-08 00Z 2.941, 06Z 2.14, 12Z 2.057, 18Z 2.799, 07-09 00Z 3.068, 06Z 2.095, 12Z 1.773
purpleair	pm25	07-06 12Z 8.076, 18Z 6.407, 07-07 00Z 6.244, 06Z 7.889, 12Z 7.974, 18Z 8.413, 07-08 00Z 7.637, 06Z 9.769, 12Z 7.773, 18Z 5.375, 07-09 00Z 5.049, 06Z 6.856, 12Z 7.626
sentinel5p	s5p_aer_ai_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_ch4_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_co_column	07-06 18Z 0.02528, 07-07 18Z 0.02495, 07-08 18Z 0.02437
sentinel5p	s5p_co_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_hcho_column	07-06 18Z 0.0002254, 07-07 18Z 0.0002152, 07-08 18Z 0.000185
sentinel5p	s5p_hcho_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_no2_column	07-06 18Z 4.086e-05, 07-07 18Z 3.867e-05, 07-08 18Z 4.307e-05
sentinel5p	s5p_no2_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_o3_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_so2_column	07-06 18Z 4.559e-05, 07-07 18Z 2.628e-06, 07-08 18Z -2.084e-05
sentinel5p	s5p_so2_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
tceq	co	07-06 12Z 0.225, 18Z 0.1882, 07-07 00Z 0.23, 06Z 0.25, 12Z 0.2167, 18Z 0.2706, 07-08 00Z 0.2, 06Z 0.2444, 12Z 0.25, 07-09 06Z 0.2222, 12Z 0.325
tceq	no2	07-06 12Z 4.619, 18Z 4.293, 07-07 00Z 3.786, 06Z 5.95, 12Z 5.672, 18Z 7.4, 07-08 00Z 6.941, 06Z 10.82, 12Z 7.497, 18Z 1.3, 07-09 06Z 5.834, 12Z 6.211
tceq	so2	07-06 12Z 0.3375, 18Z 0.075, 07-07 00Z -0.23, 06Z -0.2687, 12Z 0.3333, 18Z -0.05294, 07-08 00Z -0.2, 06Z -0.1867, 12Z -0.03889, 07-09 06Z -0.1556, 12Z -0.1

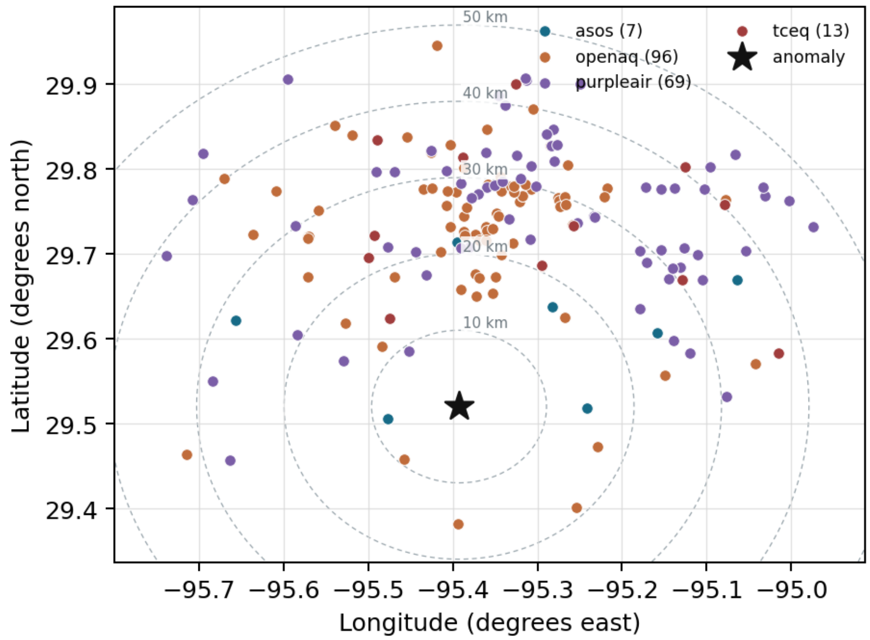
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

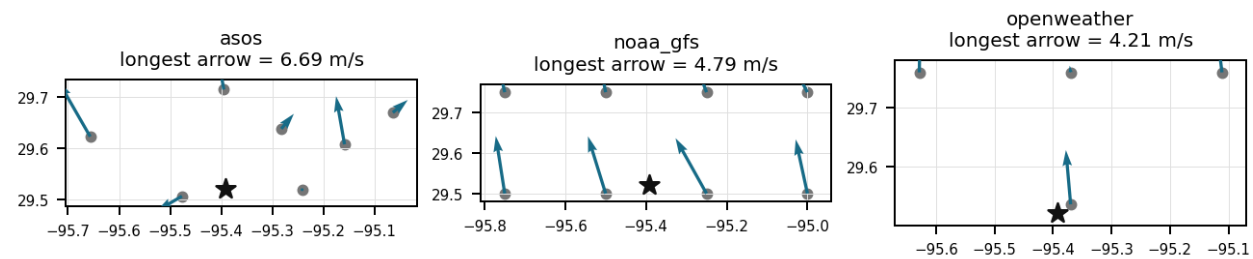
Source	Metric	Values
asos	humidity	AXH (8.318 km) 73.04, LVJ (14.592 km) 71.86, HOU (16.828 km) 72.82, MCJ (21.54 km) 72.39, EFD (24.595 km) 68.68, SGR (27.938 km) 76.43, T41 (35.81 km) 75.1
asos	temperature	AXH (8.318 km) 28.32, LVJ (14.592 km) 29.18, HOU (16.828 km) 29.35, MCJ (21.54 km) 29.02, EFD (24.595 km) 30.37, SGR (27.938 km) 28.8, T41 (35.81 km) 28.89
asos	wind_direction	AXH (8.318 km) 80.72, LVJ (14.592 km) 117.5, HOU (16.828 km) 165.1, MCJ (21.54 km) 141.1, EFD (24.595 km) 148.7, SGR (27.938 km) 149.7, T41 (35.81 km) 163.8
asos	wind_speed	AXH (8.318 km) 0.9364, LVJ (14.592 km) 1.83, HOU (16.828 km) 2.896, MCJ (21.54 km) 2.891, EFD (24.595 km) 3.213, SGR (27.938 km) 2.583, T41 (35.81 km) 2.654
noaa_gfs	gh_500	gfs:29.50,-95.50 (10.645 km) 5911, gfs:29.50,-95.25 (13.973 km) 5911, gfs:29.75,-95.50 (27.574 km) 5910, gfs:29.75,-95.25 (29.018 km) 5911, gfs:29.50,-95.75 (34.669 km) 5911, gfs:29.50,-95.00 (38.049 km) 5911, gfs:29.75,-95.75 (42.968 km) 5910, gfs:29.75,-95.00 (45.732 km) 5911
noaa_gfs	pbl_height	gfs:29.50,-95.50 (10.645 km) 684.5, gfs:29.50,-95.25 (13.973 km) 632.2, gfs:29.75,-95.50 (27.574 km) 862.3, gfs:29.75,-95.25 (29.018 km) 872.2, gfs:29.50,-95.75 (34.669 km) 672.5, gfs:29.50,-95.00 (38.049 km) 639.2, gfs:29.75,-95.75 (42.968 km) 660.3, gfs:29.75,-95.00 (45.732 km) 755.9
noaa_gfs	precipitable_water	gfs:29.50,-95.50 (10.645 km) 45.24, gfs:29.50,-95.25 (13.973 km) 46.16, gfs:29.75,-95.50 (27.574 km) 45.84, gfs:29.75,-95.25 (29.018 km) 46.69, gfs:29.50,-95.75 (34.669 km) 45.5, gfs:29.50,-95.00 (38.049 km) 46.95, gfs:29.75,-95.75 (42.968 km) 45.29, gfs:29.75,-95.00 (45.732 km) 47.24
noaa_gfs	surface_pressure	gfs:29.50,-95.50 (10.645 km) 1014, gfs:29.50,-95.25 (13.973 km) 1015, gfs:29.75,-95.50 (27.574 km) 1013, gfs:29.75,-95.25 (29.018 km) 1015, gfs:29.50,-95.75 (34.669 km) 1013, gfs:29.50,-95.00 (38.049 km) 1016, gfs:29.75,-95.75 (42.968 km) 1012, gfs:29.75,-95.00 (45.732 km) 1016
noaa_gfs	t_850	gfs:29.50,-95.50 (10.645 km) 19.02, gfs:29.50,-95.25 (13.973 km) 19.07, gfs:29.75,-95.50 (27.574 km) 19.04, gfs:29.75,-95.25 (29.018 km) 19.02, gfs:29.50,-95.75 (34.669 km) 19.02, gfs:29.50,-95.00 (38.049 km) 19.1, gfs:29.75,-95.75 (42.968 km) 19.08, gfs:29.75,-95.00 (45.732 km) 19.09
noaa_gfs	u_10m	gfs:29.50,-95.50 (10.645 km) 0.1066, gfs:29.50,-95.25 (13.973 km) 0.1358, gfs:29.75,-95.50 (27.574 km) 0.3424, gfs:29.75,-95.25 (29.018 km) 0.3008, gfs:29.50,-95.75 (34.669 km) 0.1908, gfs:29.50,-95.00 (38.049 km) 0.2049, gfs:29.75,-95.75 (42.968 km) 0.5308, gfs:29.75,-95.00 (45.732 km) 0.05326
noaa_gfs	v_10m	gfs:29.50,-95.50 (10.645 km) 2.077, gfs:29.50,-95.25 (13.973 km) 1.998, gfs:29.75,-95.50 (27.574 km) 2.361, gfs:29.75,-95.25 (29.018 km) 2.263, gfs:29.50,-95.75 (34.669 km) 2.186, gfs:29.50,-95.00 (38.049 km) 2.231, gfs:29.75,-95.75 (42.968 km) 2.505, gfs:29.75,-95.00 (45.732 km) 2.076
openaq	ozone	3055 (0.0 km) 0.01569, 312 (13.969 km) 0.01709, 320 (16.85 km) 0.01487, 325 (20.747 km) 0.005727, 311 (22.073 km) 0.01711, 3378 (27.1 km) 0.01505, 1319333 (28.091 km) 0.02153, 332 (30.475 km) 0.02062, +8 more
openaq	pm10	13380082 (27.1 km) 32.3, 15852419 (28.582 km) 28.12, 271 (30.475 km) 16.9
openaq	pm25	14152710 (9.263 km) 7.766, 13955814 (11.808 km) 3.954, 8967021 (14.63 km) 6.651, 15472101 (15.293 km) 5.223, 14140752 (15.341 km) 6.575, 11987924 (15.404 km) 2.676, 11638043 (16.705 km) 2.496, 14157975 (16.848 km) 7.538, +69 more
openweather	cloud_cover	grid:south (2.792 km) 23, grid:center (26.788 km) 24.96, grid:west (35.114 km) 20.99, grid:east (38.11 km) 23.17
openweather	humidity	grid:south (2.792 km) 74.4, grid:center (26.788 km) 75.94, grid:west (35.114 km) 75.67, grid:east (38.11 km) 78.81
openweather	precipitation	grid:south (2.792 km) 0.001944, grid:center (26.788 km) 0.08324, grid:west (35.114 km) 0.02028, grid:east (38.11 km) 0.1236
openweather	pressure	grid:south (2.792 km) 1016, grid:center (26.788 km) 1016, grid:west (35.114 km) 1016, grid:east (38.11 km) 1016
openweather	temperature	grid:south (2.792 km) 29.51, grid:center (26.788 km) 29.78, grid:west (35.114 km) 29.49, grid:east (38.11 km) 29.55
openweather	wind_direction	grid:south (2.792 km) 201.2, grid:center (26.788 km) 202.8, grid:west (35.114 km) 177.2, grid:east (38.11 km) 194.2
openweather	wind_speed	grid:south (2.792 km) 2.168, grid:center (26.788 km) 2.064, grid:west (35.114 km) 1.979, grid:east (38.11 km) 2.874
purpleair	pm25	161003 (9.186 km) 3.485, 161015 (14.492 km) 6.942, 166451 (17.628 km) 7.361, 293735 (20.756 km) 8.649, 276512 (20.799 km) 6.375, 46237 (20.895 km) 0.1208, 293725 (20.922 km) 8.28, 186315 (22.489 km) 8.058, +61 more
tceq	co	482011035 (27.078 km) 0.1196, 482011039 (30.456 km) 0.1891, 482011052 (32.704 km) 0.3946

Source	Metric	Values
tceq	no2	480391004 (0.017 km) 2.653, 482010416 (20.738 km) 4.88, 482010055 (22.069 km) 2.965, 482011066 (24.389 km) 5.238, 482011035 (27.078 km) 10.69, 482011039 (30.456 km) 4.579, 482011052 (32.704 km) 14.12, 482010047 (36.131 km) 7.811, +4 more
tceq	so2	482010051 (13.984 km) -0.1259, 482011035 (27.078 km) -0.05962, 482011039 (30.456 km) 0.04167

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

Winds shifted to a northeasterly direction (60 degrees) at speeds of approximately 3 m/s, advecting ozone from upwind industrial areas.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

A rapid collapse of the planetary boundary layer from over 930 meters to approximately 222 meters decoupled the surface layer, while light winds of around 3 m/s allowed daytime ozone to persist into the evening with delayed titration.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

Meteorology favored persistence and transport of an earlier-produced ozone plume: surface winds near the anomaly were light to moderate (~3.1 m/s) with easterly to southeasterly character nearby, and model 10 m flow around 00Z also supported onshore/southerly transport into Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

The anomalous pollutant is ozone at 29.520, -95.392, measured at 0.04 ppm on 2026-07-08T03:00:00, compared with an expected value of 0.014725584278787614 ppm; this was flagged as moderate severity by the detection methods.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

The nearest air quality monitoring station recorded a local nitrogen dioxide (NO_2) concentration of 2.9 ppb concurrent with the ozone anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

High daytime temperatures exceeding 32-34°C and significant formaldehyde and nitrogen dioxide columns on July 7 established a strong photochemical foundation for high ozone levels prior to the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

High levels of nitrogen dioxide (NO_2) and ozone (O_3) are contributing to the poor air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

The SO2 levels are not significantly different from normal, ruling out SO2 as a primary cause.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

The anomaly was detected in the vicinity of Houston, Texas, with a nearest station located approximately 26.788 km away from the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

Low local nitrogen dioxide levels of 2.9 ppb reduced nighttime titration, allowing the advected ozone to persist.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

Elevated levels of nitrogen dioxide and formaldehyde observed earlier in the day provided the necessary chemical precursors to sustain high daytime ozone production, which then persisted into the late evening.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

The presence of volatile organic compounds (VOCs) is also a possible cause, as they can react with NO_x and sunlight to form ground-level ozone.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

Local chemistry and weak removal likely helped sustain the ozone enhancement: there was little precipitation at the event time, low PM2.5 and PM10, and no strong contemporaneous indication of fresh NO titration at the nearest regulatory NO_2 site (NO_2 near 2.9 ppb at the anomaly time), which is consistent with relatively aged air where ozone can remain elevated into the evening rather than being rapidly destroyed near fresh combustion sources.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

Weather patterns may have also played a role in the air quality anomaly in Houston, Texas.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

The rapid collapse of the planetary boundary layer from over 900 meters to approximately 220 meters, combined with surface wind speeds of around 3 m/s, suggests that downward vertical mixing from the ozone-rich residual layer may have transported ozone to the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

The air quality anomaly in Houston, Texas was characterized by elevated levels of NO2, O3, and SO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

A carryover residual-layer ozone plume from earlier daytime photochemistry is a leading hypothesis: area ozone was already elevated in the 00Z–06Z window on 2026-07-08 (6 h mean about 0.025 ppm before dropping later), and the anomaly time had warm evening temperatures with a still moderately deep boundary layer near 00Z that would allow ozone formed earlier to remain aloft and mix to the surface as conditions evolved overnight.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

The anomaly occurred on July 7, 2026, at approximately 20:37 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

High levels of CO and NO2 are contributing to the air quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

Local emissions from industrial activities and vehicle traffic likely contributed to the air quality anomaly in Houston, Texas.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

A moderate nighttime ozone anomaly of 0.04 ppm was detected in Houston at 03:00 UTC on July 8, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

Northeasterly surface winds observed near the anomaly site may have horizontally advected ozone or its precursors from the highly industrial Houston Ship Channel region.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

Weather patterns such as heatwaves or stagnant air masses may be exacerbating the issue by trapping pollutants near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

There is little evidence at the anomaly time for a fresh combustion, dust, or sulfur-dominated plume driving the signal, because PM2.5 stayed low, satellite CO was not elevated, and nearby SO2 observations were near zero or negative.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

Residual daytime ozone persisting into the evening is strongly supported: the local ozone at 03:00 UTC was 0.04 ppm versus an expected 0.0147 ppm, and the regional 6 h mean ozone for 00Z on 2026-07-08 was already elevated at 0.0252 ppm before collapsing by 06Z to 0.0040 ppm, indicating the anomaly occurred during the evening decay phase rather than as an isolated spike. Warm conditions near 26 C, only partial cloud cover around 24–30%, no rain at the anomaly time, and a still moderately deep evening boundary layer near 933–1004 m are all compatible with delayed loss and downward mixing of ozone produced earlier in the day.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

Purely local fresh precursor accumulation is less favored, though not impossible: contemporaneous nearby NO_2 was low at about 2.9 ppb and SO_2 near zero or negative, TROPOMI columns do not show a strong coincident enhancement in CO, HCHO, or SO_2 before the event, and PM2.5/PM10 were low to modest, arguing against a fresh combustion or sulfur plume driving the ozone. However, overnight humidity was high and winds later weakened markedly, so some local trapping of already aged oxidants or VOC-rich air cannot be excluded.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

The anomaly is related to elevated levels of hydrogen chloride (HCl) and sulfur dioxide (SO_2).

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

At the time of the anomaly, the nearest meteorological station recorded a temperature of 26.11 degC and light winds of 3.09 m/s from an east-northeasterly direction of 60 degrees.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

At the anomaly time, ozone at the site was substantially elevated (0.04 ppm versus expected 0.0147 ppm), making this a real moderate positive evening ozone excursion rather than a small fluctuation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

Within the local monitoring context, openaq ozone 6-hour mean for 2026-07-08 00Z was 0.0252 ppm, indicating the anomaly occurred during an elevated ozone period relative to the broader multi-day mean of 0.017 ppm across nearby ozone stations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

Transport from urban/industrial precursor sources is also plausible. Near the anomaly, surface winds were modest and from the E/NE at the closest ASOS site (~60 degrees, ~3.1 m/s), while regional modeled 10 m flow around 00Z was toward the north from a southerly component, indicating directional variability consistent with plume meander/recirculation; Houston ozone commonly spikes when aged urban-industrial air is advected rather than being fully dispersed.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

A shift in wind direction toward the east-northeast near the anomaly suggests that horizontal advection of ozone-rich air or precursors from the industrial Ship Channel sector contributed to the elevated concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

An ozone anomaly of 0.04 ppm was recorded at coordinates (29.520, -95.392) on July 8, 2026, at 03:00:00 UTC, exceeding the expected baseline of approximately 0.015 ppm.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The air quality anomaly is likely caused by a combination of factors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

Transport from the Houston urban/industrial corridor is also well supported: near the anomaly, surface winds were light to modest but directional fields show a shift consistent with advection, with ASOS nearest-station winds near 60 degrees at 02:55 UTC, ASOS 00Z mean direction about 117 degrees, and gfs 10 m flow at 00Z indicating southeasterly onshore transport. Such flow can carry aged ozone or precursor-rich air from the Ship Channel/urban core toward the anomaly area while wind speeds around 2.8–3.2 m/s are sufficient for plume movement without rapid dilution.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

At nearly the same time, nearby surface observations showed about 26.1 C temperature, about 65% relative humidity, wind from about 60 degrees at about 3.09 m/s, while nearby PM2.5 was low at about 4.8 to 5.9 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.